

Stored Procedures (Self-Learning)

Q. What is a Stored Procedure Explain how to create and execute them, discuss parameters, and highlight the advantages of using stored procedures in a database.

[DEF] Definition to Remember

A Stored Procedure is a pre-compiled, named block of SQL statements and procedural logic (IF/ELSE, loops) stored directly on the database server. Applications invoke it by name using EXECUTE or CALL, rather than sending raw SQL over the network. Used in Oracle (PL/SQL), SQL Server (T-SQL), and MySQL.

1. Creating and Executing a Stored Procedure

Syntax:

```
CREATE PROCEDURE
AS
BEGIN
    --
END;
```

Execution:

```
EXECUTE
CALL P
```

2. Parameters in Procedures

Parameter Type	Direction	Description
IN	Caller -> Procedure	Passes a value into the procedure (def
OUT	Procedure -> Caller	Returns a value back to the caller
INOUT	Both directions	Accepts and modifies a value, then ret

Example:

```
CREATE PROCEDURE
AS
BEGIN
    UP
END;

-- Cal
CALL G
```

3. Advantages of Stored Procedures

Advantage	Explanation
Pre-compilation	Compiled once; subsequent calls use th
Reduced Network Traffic	Client sends one procedure call instea
Security	Users can EXECUTE a procedure without
Code Reusability	Business logic is centralized; change

[MUST-WRITE] Must-Write Points (for 10 marks)

1. Stored Procedure = named, pre-compiled block of SQL + procedural logic stored in the database.
2. Created with CREATE PROCEDURE; executed with EXECUTE or CALL.
3. Three parameter types: IN (input), OUT (output), INOUT (both).

4. Pre-compilation makes procedures faster than raw SQL queries for repeated operations.
5. Reduced Network Traffic: one call replaces hundreds of SQL lines sent over the network.
6. Security: users can execute a procedure without direct table access, preventing SQL Injection.
7. Code Reusability: centralized business logic -- update once, all applications benefit.

[RECALL] Quick Recall

Stored Procedure -> Pre-compiled -> CREATE/EXECUTE -> Parameters (IN/OUT/INOUT) -> Advantages: Faster + Reduced Network Traffic + Security + Code Reusability